Base Model

$$\frac{dp}{dt} = a\,g(t) - c\,p(t)$$

$$\frac{dg}{dt} = b\,p(t) - d\,g(t)$$

⊛ continuous Multivariate Linear Dynamical Model

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$$\frac{d\vec{x}}{dt} = A\vec{x}$$

soln: $\vec{x}(t) = e^{At}\vec{x}_0$, $\vec{x}_0 = \begin{bmatrix} P_0 \\ G_0 \end{bmatrix}$

$$\frac{d\vec{x}}{dt} = A\vec{x}$$

$$\begin{bmatrix} \frac{dp}{dt} \\ \frac{dG}{dt} \end{bmatrix} = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix} \begin{bmatrix} p \\ g \end{bmatrix}$$

Sign of eigenvalues λ_1, λ_2 of A gives tendancy of eigenvectors $\vec{v}_1$ and $\vec{v}_2$

Model soln

$\vec{x}(t) = e^{At}\vec{x}_0$, $A = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix}$, $\vec{x}_0 = \begin{bmatrix} P_0 \\ G_0 \end{bmatrix}$, $\vec{x}_t = \begin{bmatrix} P_t \\ g_t \end{bmatrix}$

Eigenvalues of $A = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix}$

Solve $\det(A - \lambda I) = 0$

$$\det \begin{bmatrix} -c-\lambda & a \\ b & -d-\lambda \end{bmatrix} = 0$$

$$(-c-\lambda)(-d-\lambda) - ab = 0$$

$$cd + c\lambda + d\lambda + \lambda^2 - ab = 0$$

$$\Rightarrow \quad \lambda^2 + (c+d)\lambda + cd - ab = 0$$

Quadratic in λ

$a = 1, \quad b = c+d, \quad c = cd - ab$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow \quad \lambda = \frac{-(c+d) \pm \sqrt{(c+d)^2 - 4(cd-ab)}}{2}$$

$\Rightarrow$ $tr(A) = -(c+d)$

$\det(A) = cd - ab$

$\Delta(A) = (c+d)^2 - 4(cd-ab)$

$= c^2 + 2cd + d^2 - 4cd + 4ab$

$\Delta(A) = c^2 - 2cd + d^2 + 4ab$

$\Delta(A) = (c-d)^2 + 4ab$

Long-Term Analysis using Tr(A) and Det(A)

- $\Delta(A) = (c-d)^2 + 4ab > 0$ always → always REAL eigenvalues
- Since $c, d > 0$, $tr(A) = -(c+d) < 0$ always.
- $det(A) = cd - ab < 0$ when $cd < ab$
- $det(A) = cd - ab > 0$ when $cd > ab$
  ⇒ no complex eigenvalues, spirals or oscillations.

Subcase I(a), I(b), I(c) not applicable. (∵ tr always negative)
Cases II and III not applicable ($\Delta(A) > 0$ always).

Subcase I(d): Line of Stable Nodes ($\Delta(A) > 0$)

- $tr(A) < 0$ ⇒ $cd = ab$
- $det(A) = 0$

Subcase I(c): Unstable Saddle Point ($\Delta(A) > 0$)

- $tr(A)$ all possible values ⇒ $cd < ab$
  ↳ always negative
- $det(A) < 0$

Subcase I(e): Stable Nodes ($\Delta(A) > 0$)

- $tr(A) < 0$ ⇒ $cd > ab$
- $det(A) > 0$